

Examining Trends of Failures and Hallucinations in AI and Automation Technologies

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ABSTRACT

Artificial intelligence and automated systems have become essential to modern society. As these technologies become more widely deployed, concerns have grown about the reliability of their outputs and the risks posed when automated or algorithmic models generate incorrect, biased, or hallucinatory results. This paper examines failures in AI and automation technologies by analyzing incident frequency, ethical issues, and legal outcomes across technology groups.

The study draws on two datasets: the AIAAIC Repository, which catalogues reported incidents involving AI, algorithmic, and automated systems, and the AI Hallucination Cases Database, which compiles legal cases involving hallucinated outputs generated by large language models (LLMs). Using Tableau Desktop, visualizations were created to examine temporal trends in reported incidents, the distribution of cases and ethical issues across system types, and the relationship between hallucination errors and judicial responses.

Overall, the findings from this paper reveal differences in how failures, ethical issues, and legal outcomes are distributed across AI and automation technologies. These differences suggest that risk levels vary depending on system design, deployment context, and use.

Keywords— *Artificial Intelligence, Technology, Generative AI, Large Language Models (LLMs), Hallucinations, Data Analysis*

I. INTRODUCTION

Over the past decade, artificial intelligence and automated systems have advanced rapidly, becoming integral to how modern society operates. Recommendation algorithms decide what users watch and consume, automation technologies have transformed industrial and manufacturing processes by improving efficiency [1], and large language models (LLMs) can now generate coherent text or respond to complex questions within seconds. As we become increasingly reliant on these technologies, concerns have intensified about the reliability of their outputs and the risks that arise when they behave unpredictably or incorrectly. These concerns form the basis of ongoing discussions about failures in AI and

automation technologies and their associated ethical issues and legal consequences.

One of the most widely recognized limitations of modern artificial intelligence systems, particularly large language models (LLMs), is the phenomenon known as *hallucination* [2]. Large language models are systems trained on vast collections of text to predict and generate language, rather than to verify factual accuracy [3]. Because their objective is to produce fluent and plausible language, these models may generate responses that sound confident, even when the information is incorrect, misleading, or entirely fabricated [4]. This behavior is commonly referred to as hallucination, where false or unsupported content is presented as if it were accurate [2]. Geoffrey Hinton, a leading figure in AI research, has warned that such systems can produce outputs that “sound confident but are wrong”, highlighting the risks of relying on AI-generated content without verification [5]. As large language models (LLMs) are increasingly used, understanding the causes and consequences of hallucinated outputs has become a growing concern.

The goal of this study is to examine patterns of failure in AI and automated systems. Using data from the AIAAIC Repository [6] and the AI Hallucination Cases Database [7], the analysis applies data visualization to examine temporal trends in reported incidents, differences in incident distribution across technology types, ethical concerns in different technology systems, and how hallucinated outputs are handled in a legal context. Together, these findings provide insight into how risks and consequences vary with system design and deployment.

The article is organized as follows: Section II describes the data collection process; Section III details the data cleaning and transformation steps; Section IV presents analysis and visualizations of the data; Section V discusses the research findings and limitations; and Section VI concludes with directions for future work. Lastly, the references are listed.

II. DATA COLLECTION

This paper features data collected from the AI Hallucination Cases Database [6] and the AIAAIC Repository

[7]. All datasets and visualizations referenced are available from the authors upon request.

Ethical Issues, External Harm's, and Internal Impacts. Figure 2.2 presents the dataset structure outlined above.

A. AI Hallucination Cases Database

The data for legal cases involving instances where generative AI systems produced hallucinated or fabricated legal content is sourced from the AI Hallucination Cases Database [6]. The dataset is publicly accessible on the project website in HTML/text format, and was downloaded on November 1, 2025, as a comma-separated values (.csv) file. The version utilized in this study is the release dated February 18, 2025, with data collected since May 8, 2023. The database remains a work in progress and is continually updated as new cases are identified.

The database comprises 532 identified legal cases (rows) and 14 variables (columns), documenting individual legal cases and their metadata. Each record represents a distinct case of AI-generated misinformation in legal proceedings or filings. The dataset includes the following key variables: *Case Name, State(s), Date, AI Tool, Nature of Hallucination,* and *Outcome.* Figure 2.1 shows the dataset structure as described above.

Case Name	State	Date	Party(ies)	AI Tool	Hallucination
Blac News David v. George Christa Farms, Inc.	N.D. California	2025-11-07	Lawyer	Implied	Fabricated case citations and misrepresented legal authority forms
Evans, et al. v. Robertson et al. (3)	USA	2025-11-07	Pro Se Litigant	Implied	Fabricated citations and false quotations (fabricated case law, false c
James Andrew Oliver v. Citibank, N.A.	D. Minnesota	2025-11-07	Pro Se Litigant	Undisputed	Misquoted citations (misrepresented case-law authorities cited in plaintiff
David Angel Silvestre, Jr. v. Christine Brothers Automotive	N.D. Michigan	2025-11-06	Pro Se Litigant	Implied	Misquoted citation (misrepresented case)
Justine Warren v. WMA, Milwaukee Truck Line, Inc.	E.D. Tennessee	2025-11-06	Lawyer	Implied	Fabricated citations (misrepresented case citations in plaintiff's
Habit Mass v. Morgan Stanley & Co. International PLC, et al.	S.D. New York	2025-11-06	Pro Se Litigant	Implied	Fabricated case citations and false quotations (apparently AI-generated
United v. City of Rio Rancho	D. New Mexico	2025-11-05	Pro Se Litigant	Implied	Misquoted citations and misrepresented authorities (AI hallucinated r
New Logan v. LHM Funding et al.	D. Utah	2025-11-05	Pro Se Litigant	Implied	Negate AI hallucinated case quotations
Buchanan v. West, Inc.	N.D. California	2025-11-05	Lawyer	Implied	Fabricated case citation and false quotations (fabricated case law an
Colman & Lewis v. PNC Bank, N.A.	D. Nevada	2025-11-05	Pro Se Litigant	Implied	Fabricated citations and false quotations of case law (fabricated case
Isabel L. Oliver v. Pity & Lerner LLP	N.D. Wisconsin	2025-11-05	Pro Se Litigant	Implied	Fabricated and misrepresented case citations (misquoted and incor
Alissa Coleman v. WFF, Jackson Investors, LLC, et al.	E.D. Wisconsin	2025-11-04	Pro Se Litigant	Implied	False, quoted (fabricated) quotations attributed to case law
Bernardo v. Frontier Heritage Communities	D. Kansas	2025-11-04	Pro Se Litigant	Implied	Misquoted citation (misrepresented case cited in plaintiff's brief)
Wheat v. Titan Team, The Money Source Inc., and Auction.com	S.D. Texas (Barackozzy)	2025-11-04	Lawyer	Undisputed	Fabricated citations and false quotes - misrepresented case law (gen
10 De 154020	CO	2025-11-03	Lawyer	Implied	Fabricated and misrepresented case citations (misquoted, doctored, r
UK v. O.K.	CO Indiana	2025-11-03	Pro Se Litigant	Implied	Misrepresented (and possibly fabricated) citations - AI-generated
Wheat v. Wale	SC New York	2025-11-03	Pro Se Litigant	Implied	Misquoted citations (non-existent cases)
104067-07-02	Israel	2025-11-02	Pro Se Litigant	Implied	none identified
Worthman v. Rysnar DAC	Israel	2025-11-01	Lawyer	Implied	Fabricated references

Fig. 1.1. The first seven columns and rows 1 to 19 of the AI Hallucination Cases Database

B. AIAAIC Repository

Incidents involving artificial intelligence, autonomous systems, algorithmic decision-making, and their associated harms were collected from the Artificial Intelligence, Algorithms, and Automation Incidents & Controversies (AIAAIC) Repository [7]. The AIAAIC Repository is a continuously updated database that compiles publicly documented cases in which AI or automated technologies have caused, contributed to, or been implicated in controversial outcomes. The dataset used in this study is the version released on November 5, 2025, with incident reports recorded since 2023.

The dataset was accessed through the AIAAIC Repository's public Google Spreadsheet and was downloaded in a comma-separated values (.csv) format. The dataset comprises 2,105 incidents (rows) and 20 variables (columns), with each row corresponding to a unique incident related to AI or autonomous systems. Key variables represent attributes of each event, including: *AIAAIC ID#, Headline, Occurred, Country(ies), Sector(s), Deployer(s), System Name(s), Technology(ies), Purpose(s), News Trigger(s), Issue(s),*

Incidents (REPORT INCIDENT)									
AIAAIC ID#	Headline	Occurred	Country(ies)	Sector(s)	Deployer(s)	Developer(s)	System name(s)	Technology(ies)	Purpose(s)
AIAAIC2113	Investigation: X algorithm amplifies right-wing, extreme content	2025	UK	Politics	Big News	xAI	X	Recommendation algorithm	Manipulate public opinion
AIAAIC2112	ChatGPT encourages Ukrainian teenager to kill himself	2025	Poland	Health	Victoria	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2111	Heavily Misused: Influential author after ChatGPT interactions	2025	USA	Health	Harriet Madsen	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2110	ChatGPT allows Jassas to win the playoffs	2025	USA	Health	Joseph Teich	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2109	ChatGPT fails to intervene in Joe Ceccardi suicide	2025	USA	Health	Joe Ceccardi	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2108	ChatGPT 'cousins' Joshua Kewling on how to commit suicide	2025	USA	Health	Joshua Kewling	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2107	Assaults Lacey commits suicide after ChatGPT relationship	2025	USA	Health	Amara Lacey	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2106	Zane Shamir commits suicide after ChatGPT encouragement	2025	USA	Health	Zane Shamir	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2105	Grokspedia under fire for AI automated fact-finding, bias, bots	2025	Canada, Czech	Media/entertainment	Wikipedia	Stevan Johansen	Conson Translatio	Bidirectional agent; Mach	Translate articles
AIAAIC2104	AI creates open-prompted Wikipedia articles in obscure language	2025	USA	Education				Deepfake	Cause disruption
AIAAIC2103	AI-generated gun video shuts down Baltimore high school	2025	USA	Education				Deepfake	Cause disruption
AIAAIC2102	Grok brushes to reap political exploit W/ Barack	2025	USA	Politics	Grok users, W/ Si aAI	Grok		Generative AI	Provide bias in inform
AIAAIC2101	Google AI system falsely call conservative activist Robby Star	2023	USA	Politics	Google users, Rod Grogg	Gemini		Generative AI	Assess model bias
AIAAIC2099	Google AI model falsely accuses US Senator Marsha Blackburn	2023	USA	Politics	Google users, Mae Grogg	Gemini		Generative AI	Large language model bias
AIAAIC2098	French teenager uses ChatGPT to plan jihadist terrorist attack	2025	France	Govt - politics; Govt		OpenAI	ChatGPT	Generative AI	Plan terror attack
AIAAIC2097	29-year-old healthcare consultant takes own life after using GPT-4	2025	USA	Health	Stephen Nuttall	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2096	13-year-old girl commits suicide after confiding in Character AI	2023	USA	Health	Juliana Parada	Character AI		Generative AI	Provide emotional support
AIAAIC2095	ChatGPT 'helps to continue meat to jump off history highway'	2025	USA	Health	Eugene Torres	OpenAI	ChatGPT	Generative AI	Provide emotional support
AIAAIC2094	Bipolar disorder sufferer ends life after bonding with ChatGPT	2025	USA	Health	Alexander Taylor	OpenAI	ChatGPT	Generative AI	Write novel
AIAAIC2093	ChatGPT persuades software developer to die in a simulation	2024	Canada	Health	Anthony Tan	OpenAI	ChatGPT	Generative AI	Machin to Collaborate on exascale
AIAAIC2092	Meta Ray-Ban glasses saves fire and homicide partner on	2025	USA	Health	Meta Ray-Ban user Miles, Elizabeth	Meta	Ray-Ban Meta	Computer vision; accou	Smart AI for fire and homicide
AIAAIC2091	Character AI tells children talk with chatbots based on Jeffrey	2025	Global	Media/entertainment	Character.AI users	Character AI		Generative AI	Provide companionship
AIAAIC2090	AI videos spread Hurricane Melissa via chatbots	2025	USA	Govt - municipal	Jane users	OpenAI		Generative AI	Slow dissemination
AIAAIC2089	Amazon replaces 14,000 jobs with AI	2025	Global	Technology	Amazon	Amazon	AI Development	Generative AI	Increase efficiency
AIAAIC2088	Microsoft's ChatGPT AI data enters linked to water shortages	2025	Mexico	Technology	Microsoft	Microsoft	CoPilot	Generative AI	Power AI systems
AIAAIC2087	Study: User dynamic pricing increases passenger fares, lowers	2023	UK	Transportation	User	User		Proting algorithm; Machine	Calculate price
AIAAIC2086	The first case: false AI video of actress Miley Cyrus	2025	India	Education	Tel Zor			Generative AI	Manipulate public opinion
AIAAIC2085	Falsified news dies by suicide after obscene AI blackmail	2025	India	Politics				Generative AI	Extend individual
AIAAIC2084	Political chatbots provide biased political advice about Dutch	2025	Netherlands	Politics	Aurachet Personen Group; Matra; C	ChatGPT; Gemini		Generative AI	Provide political advice

Fig. 2.1. The first ten columns and rows 1 to 38 of the AIAAIC Repository

III. DATA CLEANING / TRANSFORMATION

Data cleaning and transformation steps were performed to ensure the datasets could be used correctly in Tableau.

A. Cleaning AIAAIC Repository Datasets

The AIAAIC Repository [7] incident and system CSV files were cleaned for use in Tableau. The extra header row caused columns to be labeled as F1, F2, F3, and so on. To resolve this, the extra row was removed in the spreadsheet before importing the data to Tableau. Figure 3.1 shows the uncleaned dataset.

Systems (REPORT SYSTEM)										
AIAAIC ID#	System name	Released	Country	Sector	Developer	Technology type	Technique	Purpose	Ethical issue	
AIAAIC1798	Whisper	2023	USA	Multiple	OpenAI	Automated speech	Machine learning	Speech-to-text	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1797	Character AI	2022	USA	Multiple	OpenAI	Generative AI	Machine learning	Chatbots	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1796	AI Overviews	2024	UK	Multiple	Google	Generative AI	Machine learning	Search engine	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1795	Uber surge pricing	2021	Australia	Transportation	Uber	Pricing algorithm	Machine learning	Calculate surge price	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1794	Crane Automated Driving System	2016	USA	Automotive	General Motors	Self-driving system	Computer vision	Autonomous driving	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1793	Upergo	2016	Global	Media/entertainment	Google	Content moderation	Machine learning	Filter inappropriate content	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1792	CPB Crow	2016	USA	Government	Customs and Border P	Facial recognition	Machine learning	Process asylum claims	Transparency; Fairness; Accountability; Appropriation; Tr	
AIAAIC1791	Facebook Tag Suggestions	2011	USA	Media/entertainment	Facebook	Facial recognition	Machine learning	Suggest friends to tag	Privacy; Transparency	
AIAAIC1790	Drum Machine	2024	USA	Media/entertainment	Lyria AI	Generative AI	Machine learning	Generate music	Privacy; Safety	
AIAAIC1789	Jennaville	2023	USA	Media/entertainment	Suno	Generative AI	Machine learning	Generate music	Accountability; Appropriation; Tr	
AIAAIC1788	Suno	2023	USA	Media/entertainment	Suno	Generative AI	Machine learning	Generate music	Accountability; Appropriation; Tr	
AIAAIC1787	Grok	2023	USA	Media/entertainment	xAI	Generative AI	Machine learning	Generate text	Accountability; Appropriation; Tr	
AIAAIC1786	Sora	2024	Italy	USA	Media/entertainment	OpenAI	Generative AI	Machine learning	Generate video	Accountability; Appropriation; Tr
AIAAIC1785	Perplexity	2022	USA	Media/entertainment	Perplexity AI	Generative AI	Machine learning	Generate information	Accountability; Appropriation; Tr	
AIAAIC1784	BarCode	2011	USA	Health	BarCode Health	Generative AI	Machine learning	Assess and predict drug	Accountability; Appropriation; Tr	
AIAAIC1783	Inspected My AI	2023	Australia	UK; U	Media/entertainment	Snop Inc	Generative AI	Machine learning	Interact; Provide information	Transparency; Fairness; Acc
AIAAIC1782	Synthesia	2023	Australia	Health	Media/entertainment	Synthesia	Generative AI	Machine learning	Generate video	Accountability; Appropriation; Tr
AIAAIC1781	ChatGPT	2023	Croatia, Kosovo	Education	Media/entertainment	OpenAI	Generative AI	Machine learning	Generate text	Transparency; Fairness; Acc
AIAAIC1780	Amazon Alexa	2014	Germany	UK; U	Consumer goods	Amazon	Voice assistant	Speech recognition	Provide information; service	Transparency; Fairness; Acc
AIAAIC1779	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1778	Lawfare Safe City	2016	Pakistan	Police	Business	Business	Facial recognition	Computer - Strengthen law enforcement	Transparency; Fairness; Acc	
AIAAIC1777	Adobe Firefly	2023	USA	Media/entertainment	Adobe	Generative AI	Machine learning	Generate images	Accountability; Appropriation; Tr	
AIAAIC1776	Sale Ranks Incident Detection Case	2023	India	Transport	Katrina; NVIDIA; Tera	CCTV; Computer vision	Deep learning	Monitor traffic violations	Transparency; Fairness; Acc	
AIAAIC1775	Tesla Autopilot, Full-Drive Driving	2014	USA	Automotive	Tesla	Driver assistance	Computer vision	Machine vision	Automate steering, accelera	Transparency; Fairness; Acc
AIAAIC1774	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1773	ChatGPT	2023	USA	Global	Multiple	OpenAI	Generative AI	Machine learning	NLP-based AI	Transparency; Fairness; Acc
AIAAIC1772	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1771	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1770	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1769	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1768	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1767	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1766	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1765	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1764	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1763	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1762	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1761	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1760	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1759	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1758	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1757	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1756	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1755	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1754	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1753	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1752	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1751	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1750	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1749	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1748	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1747	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1746	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1745	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1744	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1743	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1742	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1741	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1740	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1739	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1738	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn Predict	Transparency; Fairness; Acc
AIAAIC1737	AI Predict	2013	USA	Banking/finance	Wells Fargo	AI Predict	Machine learning	AI Predict analysis	Deep learn	Transparency; Fairness; Acc

Formula

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IF CONTAINS(LOWER([Ethical issue]), "privacy")
OR CONTAINS(LOWER([Ethical issue]), "surveillance") THEN
"Privacy & Surveillance"

ELSEIF CONTAINS(LOWER([Ethical issue]), "bias")
OR CONTAINS(LOWER([Ethical issue]), "fairness")
OR CONTAINS(LOWER([Ethical issue]), "discrimination") THEN
"Fairness & Bias"

ELSEIF CONTAINS(LOWER([Ethical issue]), "accuracy")
OR CONTAINS(LOWER([Ethical issue]), "reliability")
OR CONTAINS(LOWER([Ethical issue]), "hallucination")
OR CONTAINS(LOWER([Ethical issue]), "misinformation") THEN
"Accuracy & Reliability"

ELSEIF CONTAINS(LOWER([Ethical issue]), "accountability")
OR CONTAINS(LOWER([Ethical issue]), "transparency")
OR CONTAINS(LOWER([Ethical issue]), "explainability") THEN
"Accountability & Transparency"

ELSE
"Accuracy & Reliability"
END

```

Fig. 4.1. Calculated field used to group ethical issue labels in the AIAAIC Repository

Calculated fields were used when variables followed consistent naming patterns. For example, in the AIAAIC Repository [7] systems sheet, ethical issue labels often contained multiple descriptors separated by semicolons. (e.g., *Privacy; Security; Transparency*). As shown in Figure 4.1, a calculated field was created to identify keywords within each label and assign each system to a simplified ethical issue group.

Groups:

- > Other or Unspecified Inaccuracies
- > Fabricated or False Citations
- ✓ Non-existent Legal Cases or Authorities
 - 1 fake case + fabricated factual allegations.
 - 16 non-existent court cases, and fabricated quotes
 - 19 fabricated Constitutional Court decisions
 - 2 fake cases + fabricated quotations attributed to real cases
 - 2 Fake Cases, plus flawed summaries
 - 3 entirely fictitious cases
 - 3 fake cases
 - 3 miscited cases and 4 non-existent ones
 - 5 non-existent cases and misrepresentation of three others
 - 6+ Fake Cases, Quotes, Citations; Fake Opinions

Fig. 5.1. Manual grouping of hallucination descriptions into type of hallucination categories

Manual grouping was applied when variables contained highly specific and long descriptors that could not be reliably grouped using calculated fields. For example, in the AI Hallucination Cases Database [6], the *Types of Hallucinations* field included detailed labels describing specific legal contexts and AI behaviors. As shown in Figure 5.1, these detailed descriptions were manually reviewed and grouped into a smaller set of hallucination categories.

IV. DATA ANALYSIS

This section examines temporal trends in reported incidents, the distribution of incidents and ethical issues across technology groups, and how hallucination-related errors are addressed in legal settings. Together, these results provide insight into how risks and consequences vary across different AI and automation technologies.

All visualizations used in this study were created in Tableau Desktop, and interactive dashboard versions are available upon request.

A. Growth of AI-Related Incidents Over Time

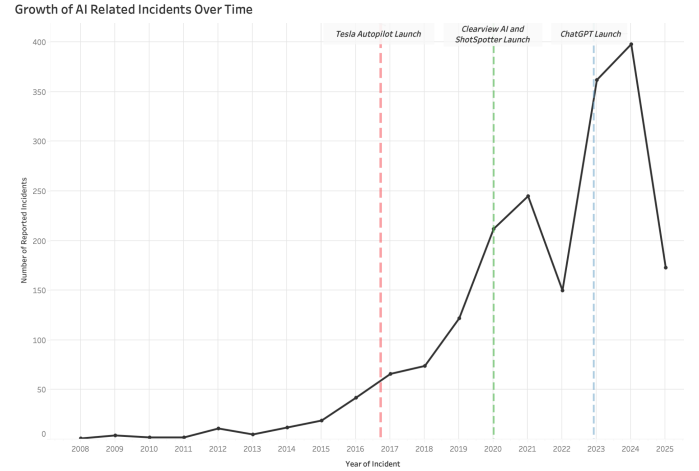


Fig. 6.1. Annual AI-related incidents with reference lines for major system releases in 2016, 2020, 2022

Figure 6.1 presents a line graph illustrating the annual number of AI-related incidents documented in the AIAAIC Repository between 2008 and 2025. The x-axis represents the year each incident occurred, while the y-axis shows the number of reported cases. The three vertical reference lines (red, green, blue) mark major technological milestones associated with visible shifts in the trend: the launch of Tesla Autopilot (2016), the launch of Clearview AI, ShotSpotter, and related facial recognition deployments (2020), and the launch of ChatGPT (2022). These reference points provide time context for interpreting changes in incident counts following major deployments of AI and automation systems.

After each technological milestone, the number of reported incidents increases noticeably, suggesting that incident growth is closely linked to wider deployment and public use of AI and automation systems. As these systems become more accessible and integrated into daily life, failures are more likely to be noticed and documented. This pattern indicates that rising incident counts reflect increased system exposure and public use.

Figure 6.1 shows a substantial increase in AI-related incidents over time. The early period of the dataset, from 2008 to 2015, shows relatively low and stable incident counts. A noticeable rise begins near the 2016 red reference line, which aligns with the introduction and growing adoption of Tesla

Autopilot [8]. Within the dataset, Tesla Autopilot accounts for 44 documented incidents, making autonomous driving and driver-assistance systems the dominant source of reported harms during this period. This phase marks the first major shift in the dataset, during which incidents involving autonomous driving and driver-assistance technologies accounted for a large share of reported cases.

A second shift occurs around 2020, near the green reference line, marking the increased adoption of algorithmic surveillance systems such as Clearview AI and ShotSpotter. Clearview AI is a facial recognition platform that extracts and matches images from large online datasets [9], while ShotSpotter is a gunshot detection system widely used by U.S. law enforcement agencies [10]. The AIAAIC Repository records a sharp increase in incidents, rising from 122 cases in 2019 to 212 cases in 2020. During this period, facial recognition and surveillance systems make up a larger portion of reported incidents, indicating a shift away from autonomous driving and toward recognition and surveillance technologies.

The most dramatic increase in documented incidents appears after 2022, which marks the launch of ChatGPT [11]. While 2022 shows a lower count of 150 incidents, the following years show the largest spikes in the figure, with 362 incidents in 2023 and 398 incidents in 2024. The sharp upward trajectory following the 2023 blue reference line aligns with the rapid adoption of generative AI systems, as large language models (LLMs) such as ChatGPT and Google Gemini became widely used by the public. During this period, a large share of incidents involved large language models (LLMs) producing misinformation, hallucinated content, or unsafe advice, contributing to the high number of documented cases in these years.

The temporary decline in documented incidents in 2022 may be partly attributed to the effects of the COVID-19 pandemic. The pandemic has disrupted normal reporting processes and slowed investigations and publications, leading to delays or gaps in documentation [12]. As a result, the lower case count in 2022 may reflect changes in reporting practices rather than an actual decline in AI or automation-related harms.

The lower number of incidents recorded for 2025 reflects incomplete annual reporting. Overall, Figure 6.1 suggests that the growth of AI and automation system incidents can be understood in three major technological phases: autonomous and driver-assistance systems (2016), algorithmic surveillance technologies (2020), and generative AI models (2022). Together, these phases suggest that shifts in incident volume are closely linked to changes in the types of technology systems being deployed and the scale of their real-world use.

B. Distribution of Incidents by Technology Groups

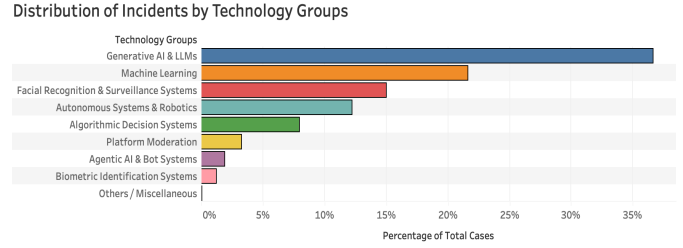


Fig. 7.1. Percentage of incident cases linked to each AI, automated, or algorithmic system type

Figure 7.1 presents a horizontal bar graph illustrating the percentage of documented incidents associated with each major technology group in the AIAAIC Repository [7]. Because the raw dataset contained over 300 highly specific and overlapping technology descriptors, these entries were consolidated into broader groups to support clearer comparisons. This consolidation involved reviewing the functional similarities among system labels and grouping them into nine higher-level domains: *Generative AI & LLMs*, *Machine Learning*, *Facial Recognition & Surveillance Systems*, *Autonomous Systems & Robotics*, *Algorithmic Decision Systems*, *Platform Moderation*, *Agentic AI & Bot Systems*, *Biometric Identification Systems*, and *Others/Miscellaneous*.

The x-axis represents the percentage of all documented cases attributed to each technology group, while the y-axis lists the nine consolidated categories. Each bar represents the proportion of incidents in the dataset associated with that specific type of system.

The distribution shows that incidents are concentrated within a relatively small number of technology categories. Generative AI & LLMs (36%), Machine Learning (22%), and Facial and Surveillance Systems (15%) together account for more than 70% of all documented cases, suggesting that systems with broad public deployment, frequent user interaction, or direct involvement in information processing or monitoring activities tend to generate a larger share of reported incidents.

A second tier of technologies, including Autonomous Systems and Robotics (13%) and Algorithmic Decision Systems (8%), also contributes to a notable portion of cases. These systems are often used in contexts such as automated driving, drone operations, and algorithmic classification tasks, which may increase the likelihood that failures become publicly documented.

Lower-frequency categories, such as Platform Moderation, Agentic AI & Bot Systems, and Biometric Identification Systems, appear far less often in the dataset. This may reflect more limited uses, fewer identified failures, or reduced visibility rather than a lower inherent risk.

The Other / Miscellaneous category represents a limited proportion of incidents, and includes entries labelled “Null”, along with systems that could not be clearly classified.

Overall, Figure 7.1 indicates that reported incidents are unevenly distributed across technological domains, with a clustering around systems that are widely deployed and operate at scale. The prominence of generative AI, machine learning, and surveillance technologies reflects their widespread use in content generation, predictive modeling, and monitoring applications. In these contexts, system outputs are more visible, making errors more likely to be noticed and reported. In contrast, technologies that are more specialized, less visible, or deployed in narrower contexts appear less often in public incident records, suggesting that differences in exposure, usage scale, and visibility contribute to differences in reporting frequency.

C. Distribution of Ethical Issues By Technology Groups

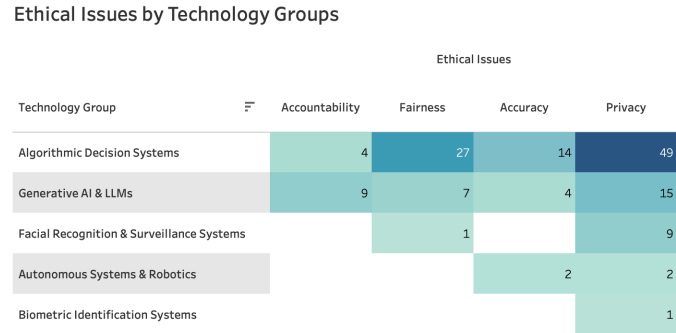


Fig. 8.1. Distribution of documented systems by ethical issue and technology group out of 130 systems

Figure 8.1 presents a heatmap showing the distribution of ethical issues across different technology groups in the AIAAIC Repository [7]. The rows represent technology groups, while columns represent ethical issue categories, including *Accountability*, *Fairness*, *Accuracy*, and *Privacy*. Each cell shows the number of documented systems within a given technology group that are associated with a specific ethical issue. Darker shading indicates higher counts of associated systems, while lighter shading represents lower counts. This visualization highlights how ethical concerns vary across technology groups and which issues are most commonly linked to each type of system.

In this analysis, *Accountability* refers to cases where it is unclear who should be blamed or held responsible when an AI system causes harm or makes a wrong decision. *Fairness* includes problems such as bias or discrimination that affect certain groups more than others. *Accuracy* refers to systems producing incorrect, misleading, or unreliable outputs. *Privacy* involves the collection or use of personal information without proper consent or protection. These categories help explain the different types of ethical problems linked to each technology group.

The distribution in Figure 8.1 shows that ethical issues are unevenly spread across technology groups. Algorithmic decision systems are strongly associated with *Fairness* and *Privacy* concerns. In the AIAAIC Repository, this category includes systems that automate or support institutional

decision-making through scoring, ranking, or classification processes [13]. These systems are commonly used to inform decisions that affect individuals, such as hiring, credit approval, risk assessment, and access to services [13]. Because such systems rely on historical data and predefined decision rules, they can reproduce or amplify existing social biases, leading to unfair or discriminatory outcomes. Privacy concerns also arise because these systems frequently process sensitive information, increasing the risk of inappropriate data use or exposure.

Similarly, generative AI & LLMs are also frequently associated with *Privacy* concerns. Privacy issues are prominent in this technology group due to how these systems are trained and used. These models are trained on large-scale datasets that may include personal or sensitive information collected from online sources. In addition, users may input private details when interacting with AI chatbots without a clear understanding of how those inputs are stored and reused. As a result, Generative AI & LLMs raise privacy concerns related to data exposure, unauthorized data use, and insufficient user control over leaked personal information.

Ethical issues associated with facial recognition and surveillance systems are concentrated around *Privacy* concerns, reflecting their use of personal data such as facial images and location information [13]. In contrast, autonomous systems and robotics, as well as biometric identification systems, appear in a relatively small number of documented systems in the dataset. Because these groups are represented by fewer systems, fewer ethical issues are observed for them. Their lower count likely reflects limited reporting in the AIAAIC Repository, rather than the absence of ethical risk.

Overall, Figure 8.1 shows that ethical issues vary by technology group and reflect differences in system use. *Privacy* concerns appear across all groups, while *fairness* and *accuracy* concerns are more concentrated in algorithmic decision systems and generative AI & LLMs. Because some groups include relatively few documented systems, these findings should be interpreted cautiously and should not be treated as generalizable conclusions.

D. AI Hallucination Types and Associated Legal Outcomes

This section analyzes the relationship between types of AI hallucinations and associated legal outcomes. Building on Figures 9.1 and 10.1, it examines how different hallucination categories correspond to patterns in judicial responses, focusing on how courts address AI-generated errors when they appear in legal proceedings.

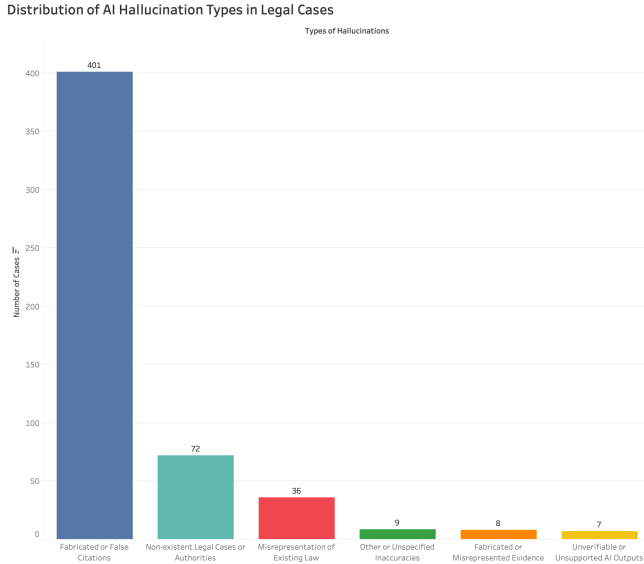


Fig. 9.1. Documented legal cases categorized by AI hallucination type

Figure 9.1 presents a bar graph illustrating the distribution of documented legal cases by type of AI hallucination, using cases from the AI Hallucination Cases Database [6]. The x-axis displays consolidated hallucination categories, while the y-axis represents the number of associated legal cases. Because the original dataset contained highly detailed and overlapping descriptions of hallucinated behaviors, individual case descriptions were reviewed and grouped into broader categories of hallucinations, including *Fabricated or False Citations*, *Non-existent Legal Cases or Authorities*, *Misrepresentation of Existing Law*, *Fabricated or Misrepresented Evidence*, *Unverified or Unsupported AI Outputs*, and *Other or Unspecified Inaccuracies*. Following this grouping process, each bar reflects the total number of documented cases associated with a hallucination type.

The distribution in Figure 9.1 shows a strong concentration of documented legal cases involving citation-related AI hallucinations. *Fabricated or False Citations* account for the largest share by a wide margin, with 401 documented cases, followed by *Non-existent Legal Cases or Authorities* with 72 cases. In contrast, *Misrepresentation of Existing Law* occurs in 36 cases, while categories such as *Fabricated or Misrepresented Evidence* (8 cases), *Other or Unspecified Inaccuracies* (9 cases), and *Unverified or Unsupported AI Outputs* (7 cases) are relatively rare.

This pattern indicates that hallucinations affecting legal authority, particularly fabricated citations and nonexistent cases, are far more likely to be identified and documented than other types of AI-generated errors. In legal proceedings, where

source verification and citation are central, fabricated or nonexistent citations are easily identified and directly undermine the credibility of legal filings [14]. As a result, citation-based hallucinations are more visible in court records and appear more frequently in documented legal decisions.

Building on the distribution of hallucination types identified in Figure 9.1, the analysis next examines how these errors are addressed once they appear in legal proceedings. Figure 10.1 analyzes the legal outcomes associated with each hallucination category, showing how different forms of AI-generated errors correspond to different judicial responses.

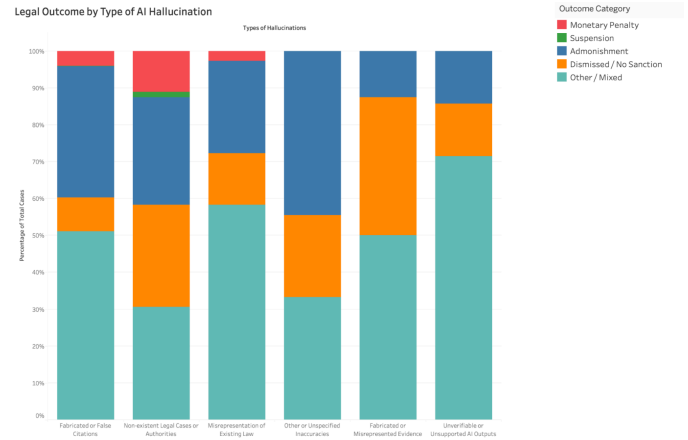


Fig. 10.1. Distribution of legal case outcomes across different AI hallucination types

Figure 10.1 presents a 100% stacked bar graph illustrating the distribution of legal case outcomes across different AI hallucination types. The x-axis lists the consolidated hallucination categories, while the y-axis represents the proportion of different legal outcomes for each hallucination type. Each bar is segmented by different legal outcomes, including monetary penalties, suspensions, admonishments, dismissals, or no-sanction cases, and other or mixed outcomes. This visualization highlights differences in the severity of legal outcomes associated with different types of AI hallucinations.

The distribution indicates substantial differences in legal outcomes across different AI hallucination types, with citation-related hallucinations associated with more severe judicial responses. In cases involving *Fabricated or False Citations*, approximately 35% result in admonishments, with an additional 4% leading to monetary penalties. Dismissals or no-sanction outcomes only account for around 10% of the cases. A similar pattern is observed for *Non-existent Legal Cases or Authorities*, where admonishments comprise approximately 30% of cases, and roughly 1% lead to suspension. Because these hallucination errors fabricate legal sources, they directly compromise the reliability of legal submissions, prompting courts to impose more severe outcomes. Together, these patterns indicate that hallucination errors affecting legal authority are more likely to result in severe judicial outcomes, such as admonishments and

monetary penalties, rather than dismissals or no-sanction decisions.

In comparison, hallucinations involving the *Misrepresentation of Existing Law* are associated with intermediate judicial responses. Outcomes are distributed across other or mixed outcomes (58%), admonishments (25%), dismissals or no-sanction decisions (14%), and monetary penalties (3%). This pattern suggests that errors involving misapplication of existing legal principles are treated more seriously than unsupported statements, but less severely than citation-based hallucinations.

The least severe judicial responses are associated with hallucination types involving unsupported or misleading content. *Unverifiable or Unsupported AI Outputs* are dominated by other or mixed outcomes, accounting for approximately 70% of cases, while admonishments comprise only about 15%. *Fabricated or Misrepresented Evidence* shows a similar pattern, with roughly half of the cases resulting in other or mixed outcomes, approximately 37% as dismissed or no sanction, and only around 13% leading to admonishments. Neither category is associated with monetary penalties or suspensions, and admonishments occur far less frequently than in citation-related hallucinations. Together, this pattern suggests that courts are less likely to impose formal sanctions when AI-generated errors involve unsupported statements, rather than fabricated legal authority.

Across all documented legal cases involving AI hallucinations, the majority of outcomes are low in severity. In total, dismissals, no-sanction decisions, and other or mixed outcomes together account for over two-thirds of all cases. This distribution indicates that courts most often address AI-related errors by resolving or acknowledging them, without imposing formal punishments.

Admonishment outcomes represent a smaller share of outcomes, appearing in approximately 20-25% of cases. This indicates that courts sometimes respond to hallucination errors by issuing formal warnings, without escalating to more severe punishments. In contrast, monetary penalties and suspensions are rare and observed only in a few hallucination types, such as *Fabricated or False Citations*, *Non-existent Legal Cases or Authorities*, and *Misrepresentation of Existing Law*.

Overall, Figure 10.1 shows that the legal consequences of AI hallucinations are unevenly distributed across cases. Most documented incidents resolve without formal punishment, with dismissals, no-sanction decisions, and other or mixed resolutions accounting for the majority of outcomes. More severe responses appear primarily in citation-related hallucination cases, indicating that judicial concern is concentrated where hallucinations affect the validity of legal sources and citations. Importantly, the number of cases varies substantially across hallucination categories, with several categories represented by relatively few observations. As a result, patterns observed for less frequent hallucination types should be interpreted cautiously, as they may not show statistically meaningful differences.

E. Legal Case Outcomes by AI Tool Attribution

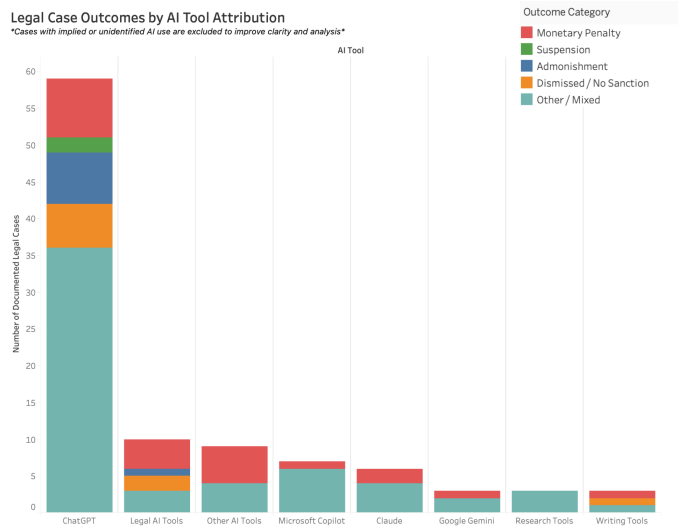


Fig. 11.1 Documented legal case outcomes across different AI tool attribution

Figure 11.4 presents a stacked bar graph comparing documented legal case outcomes across AI tools with explicit attribution, using cases drawn from the AI Hallucination Cases Database [6]. To improve interpretability and reduce inconsistencies from overlapping tool labels, individual systems were consolidated into broader AI tool groups, including *ChatGPT*, *Microsoft Copilot*, *Google Gemini*, *Claude*, *Legal AI Tools*, *Writing Tools*, *Research Tools*, and *Other AI Tools*. The x-axis displays these grouped tools in descending order by total number of documented cases, while the y-axis represents the number of associated legal cases. Each bar is segmented by outcome category, including monetary penalties, suspensions, admonishments, dismissed or no-sanction cases, and other or mixed outcomes. Cases in which AI use was implied or identified but not specifically attributed to a named system were excluded.

The distribution shown in Figure 11.4 indicates that legal cases involving AI hallucinations are unevenly distributed across tools, with cases concentrated among a small number of widely deployed systems. Cases attributed to ChatGPT represent the largest share of documented outcomes and appear across all outcome categories in the figure, including dismissals, admonishments, suspensions, and monetary penalties. This pattern likely reflects ChatGPT’s widespread adoption and frequent use in legal-related tasks, such as drafting, research, and citation generation, where hallucinated outputs may be incorporated into legal filings, making errors more likely to be identified during judicial review. At the same time, the prominence of ChatGPT in this dataset may also be influenced by reporting and attribution effects, as more widely used and publicly recognized systems are more likely to be identified by courts when AI involvement is documented.

In contrast, other generative systems, including Microsoft Copilot, Claude, and Google Gemini, appear less frequently in legal records and are associated with a more limited set of outcomes. While this difference does not imply a lower risk, it suggests that variations in adoption, visibility, and how courts

identify and name AI tools play a significant role in shaping which AI tools appear most frequently in documented legal cases.

Figure 11.1 also reveals variation in how courts respond once different AI tools are implicated. Legal-specific AI systems, including CoCounsel, Lexis+AI, Ghostwriter Legal, Westlaw Precision, and Legal Genius, appear in relatively few documented cases but are more often associated with formal court responses such as admonishments and monetary penalties, rather than dismissals or no-sanction outcomes. These systems are large language models (LLMs), developed specifically for legal tasks such as legal research, citation, document drafting, and case analysis [15]. Because their outputs inform legal reasoning and court materials, errors such as hallucinated citations and incorrect legal claims are more likely to be identified during judicial review. Importantly, this pattern does not imply that legal-specific AI systems are inherently more error-prone; rather, it suggests that the legal contexts in which they are applied increase the likelihood that errors will result in documented legal consequences.

Furthermore, Figure 11.1 shows that cases involving research tools and writing tools appear infrequently in documented legal cases and are associated with limited legal consequences. Research tools, which include entries such as Google Search and Google Scholar where AI involvement was alleged, are exclusively associated with other or mixed outcomes, and do not exhibit severe court responses. Writing tools, including Grammarly, ProWritingAid, and MobiOffice, likewise appear in a small number of cases, and are linked to a range of outcomes, including dismissals, other or mixed outcomes, and a single instance of monetary penalty. Unlike large language models (LLMs) used for legal reasoning and drafting, these tools are typically used for minor assistance, such as language editing and paraphrasing [16]. As a result, errors associated with research and writing tools are less likely to affect legal arguments, which may explain why their involvement results in fewer and less severe judicial responses.

Overall, Figure 11.1 demonstrates that legal outcomes associated with AI hallucinations vary across tools, reflecting differences in how these systems are used and the contexts in which their outputs appear. Taken together, the findings suggest that variation in legal outcomes reflects differences in usage context rather than differences in the AI tools themselves.

A. Discussion

Across all analyses, the results show that failures, ethical issues, and legal consequences associated with AI and automation systems are unevenly distributed. Temporal trends indicate that reported failures increase sharply following the widespread deployment of major systems, including autonomous driving technologies, algorithmic surveillance tools, and generative AI models. This suggests that incident growth is closely linked to system exposure, the scale of its use, and public interaction.

The distribution of incidents by technology groups further supports this interpretation. A small number of widely deployed technologies, particularly generative AI and large language models (LLMs), machine learning systems, and facial recognition and surveillance technologies, account for the majority of documented incidents. These systems are commonly used in public, where errors are more visible and likely to be documented. In contrast, more specialized or narrowly deployed technologies appear less frequently in public incident records, likely reflecting differences in usage scope and reporting visibility.

Analysis of ethical issues reveals that concerns such as privacy, fairness, accuracy, and accountability align closely with how systems function. Privacy concerns appear across all technology groups, reflecting the widespread reliance on modern technologies and automated systems, large-scale data collection and processing [17]. In contrast, fairness and accuracy issues are more concentrated in systems that automate decisions or content generation systems, such as algorithmic decision systems and large language models. These patterns suggest that ethical risk is shaped more by the technology’s roles, its data dependency, and decision-making authority.

The legal analysis of hallucination-related cases further highlights how context influences consequences. Courts respond most strongly to hallucinations that undermine legal authority, particularly fabricated citations and non-existent cases, while unsupported or misleading content is more often resolved without formal sanctions. This indicates that legal risk mainly arises when AI-generated hallucination errors directly affect the validity of legal documents or court proceedings. Overall, the findings suggest that risks from AI failures are not the same across all systems, but depend on how and where technology is used.

B. Limitations

Several limitations affect the interpretation of these results. First, both datasets, the Hallucination Cases Database [6] and the AIAAIC Repository [7], rely on publicly documented incidents and legal cases. As a result, systems that are widely deployed, highly visible, or more popular are more likely to appear more frequently, while incidents with less visible systems may go unreported. This reporting bias may influence which technologies appear most frequently in the analysis. For example, technology groups are not equally represented in the datasets. Some groups, such as generative AI and surveillance systems, include many documented systems or cases, while

others contain relatively few. This imbalance limits the extent to which patterns can be directly compared across all technology groups.

Second, there are limited sample sizes in some analysis sections. Certain parts of the analysis, particularly those based on the systems sheets and legal cases, involve a relatively small number of observations. For example, in Figure 9.1, there were up to 401 samples for fabricated or false citations, while there were only 8 samples for fabricated or misrepresented evidence. As a result, the findings in this section may only represent observable tendencies in the available data, rather than broad, general conclusions.

Third, the analysis identifies patterns and associations between system types, ethical issues, and legal outcomes, but it does not establish causal relationships. For example, in Figure 6.1, increases in reporting incidents following major system deployments reflect higher exposure and reporting, not direct evidence that these systems caused more failures.

VI. CONCLUSION & FUTURE WORK

This paper examined failures, ethical issues, and legal consequences associated with AI and automation systems using data from the AI Hallucination Cases Database [6] and the AIAAIC Repository [7]. The results show that failures and risks are unevenly distributed across technologies and are concentrated in systems that are widely deployed, highly visible, and have high public interaction. Incident trends increase following major technology deployments, indicating that system exposure and scale of use play a major role in reported failures. Ethical concerns vary by system function, with privacy appearing across most technologies, and fairness and accuracy concentrated in decision-making and generative systems. Legal outcomes further show that courts respond most strongly to hallucination errors that undermine legal authority, while other errors are often resolved without formal sanctions. Overall, the findings suggest that risks from AI failures depend on system design, deployment context, and how they are used.

Future work could extend this analysis by incorporating more complete and standardized reporting of AI incidents, particularly for less visible or specialized systems. Additional variables, such as deployment context and severity of harms, could allow for better comparisons across technologies. The authors of this paper also suggest that further research could examine hallucinogens in domains beyond law, as well as conditions and practices under which hallucinations are mitigated to support safe AI use.

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